YouTube: The Adpocalypse

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Abstract— This paper seeks to address the recent changes to YouTube's monetary policy, how it affects ad revenue for YouTube channels and what I propose YouTube should change to improve their monetary policy. At the beginning of this paper, I shall introduce certain terms relevant to YouTube, explain how YouTube works and put the current situation called the "YouTube Adpocalypse" into perspective.

I. Introduction (*What is YouTube?*)

YouTube is a free video sharing website owned by Google Inc., that makes it easy to find and upload online videos. The site was originally created in 2005, and over the years has greatly picked up in popularity with over 6 billion viewing hours recorded every month. There are a variety of YouTube videos including, but not limited to, tutorials, cooking demos, science lessons, adorable pets and music videos. On the site, users can create their own profile and engage in watching, liking, sharing, commenting and upload their own videos to their YouTube channel. The ability to like a video raises its popularity up when a user searches for a tag related to the video. YouTube channels often ask viewers to like a video during their outro screen (the ending screen of a video). Here is a full list of the main functions available on YouTube-

1. Users can search for and watch videos
2. Create a personal YouTube channel
3. Upload videos to your channel
4. Like/Comment/share other YouTube videos
5. Users can subscribe/follow other YouTube channels and users
6. Create playlists to organize videos and group videos together

To create a YouTube account, users must be 13 years old plus and have a Google account to login with. With over a billion users and 300 hours of footage uploaded per min, YouTube's demographic is extremely diverse thus videos having a rating system. This allows YouTube to protect children from inappropriate adult material. In addition to this, YouTube channels can restrict who watches their videos with YouTube's privacy option. YouTube also has a safety mode, which is a setting that allows users to censor out all mature content from their home page and searches [13].

YouTube channels thrive on their subscribers, people who will receive notifications about uploaded videos of their subscriptions. The activity a YouTube channel receives is primarily dependent on the number of subscribers a channel has. The more subscribers mean the more consistently active viewers a channel will have and thus increase the popularity of the channel in YouTube's search algorithm. When viewers watch a video, they may receive an ad which contributes to the YouTuber's source of income provided they have been allowed to monetize their videos. YouTube controls this flow of ad revenue with a software called Google AdSense, which YouTube channels connect their channel to. They must first fulfill certain requirements to be allowed to monetize their videos. Many large companies will promote their products on YouTube channels in return for monetary compensation. Many large companies will promote their products on YouTube channels in return for monetary compensation. The details of YouTube monetization are listed below and taken from 2015 YouTube policies.

Once they link Google AdSense to their channel, they can make 68% of the ad revenue earned from their videos. YouTube starts charging advertisers once a viewer has watched 30 seconds or half of an ad, and mostly charges $.18 per view. YouTuber's can also display sponsored products in their videos to earn money, typically around $10,000 per 100,000 views. Many viewers skip the ads, so on average 15% of the viewers are counted as paid viewers. If a video has 2,000 views and 15% of the viewers watch the ad, then there would be 300 paid views. With the $.18 per view rate, this would mean the video would net an income of $54. However, the content creator receives only 68% of that amount ($36.72). The top and most popular YouTubers can earn millions of dollars. In 2014, the number one most subscribed YouTuber named "PewDiePie" earned $4 million with over 25 million subscribers. There are two types of advertisements on YouTube- Cost Per Click (CPC) and Cost Per View (CPV). CPC is, as its name implies, when a YouTuber is paid money based on clicks. If someone clicks on an ad in a video, typically in a square banner, then the advertiser will be charged. On the other hand, CPV is when an advertiser pays a YouTuber money based on the number of views their videos receive. If someone has watched at least 30 seconds or half of an ad, the advertiser gets charged once for that viewer [12].

Recently, YouTube started to change the way revenue is earned from advertising through their YouTube Partner Program (YPP). In April of 2017, they set an eligibility requirement of 10,000 lifetime views for the YPP. This enforcement became even stricter beginning January 2018, with a requirement of 4,000 hours of watch time within the past 12 months and at least 1,000 subscribers [9]. YouTube enforced these changes to dissuade bad content creators from being able to monetize their videos. Certain extremist videos received spotlight from becoming monetized and YouTube decided to tighten its policies on monetization. While the intent is understandable, these changes make it even more difficult for new content creators to grow their channels in an economic manner [10].

These changes have also dramatically impacted the ad revenue earned for YouTubers like Felix 'PewDiePie'

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Kjellberg, Phillip DeFranco, and h3h3productions. Advertisers can now opt out of monetizing videos that contain sensitive subjects such as “tragedy and conflict”, “sensitive social issues”, “sexually suggestive content”, “sensational and shocking” and “profanity & rough language”. With such limited options, YouTube videos have become very difficult to monetize each after becoming a part of the YPP. If a video contains a swear word it in (majority of them do), then it will be filtered out of advertisers list of videos to monetize and be deemed Not Advertiser-Friendly”. This has led to the uproar known as the “Adpocalypse”.

Fig. 1. PewDiePie "DEATH TO ALL JEWS" video scares advertisers.

YouTube is unwilling to back down from their new policies, due to the pressure from advertisers that do not want their brands supporting malicious content creators. Thus, I propose a change to monetization on YouTube that I believe will succeed based on its popularity on a similar platform. I think YouTube should take note from Amazon’s live video streaming platform called Twitch.tv, also known as Twitch. They should take inspiration from Twitch’s monetization policies and implement a donation feature for uploaded videos. On Twitch, content creators can receive direct transactions from viewers that opt to donate to them and earn a sizeable income. Twitch’s popularity has continued to increase, especially among YouTuber’s trying to survive the “Adpocalypse”. If YouTube implements a donation feature, I believe much of the backlash from their new policies will cease and YouTuber’s will be able to earn income of their channels once again.

II. CHANGING YOUTUBE MONETIZATION (*IMPLEMENTING DONATIONS*)

A. Is the change necessary? (Data analysis)

Before going into more detail about the suggested donation feature, let us first look at whether the change is necessary.

During the “Adpocalypse” advertisers including AT&T, Verizon, Dish Network and PepsiCo discontinued their services with Google’s non-search advertisements (i.e. YouTube) [1]. An AT&T spokesman gave the following statement regarding YouTube, “We are deeply concerned that our ads may have appeared alongside YouTube content promoting terrorism and hate. Until Google can ensure this won’t happen again, we are removing our ads from Google’s non-search platforms.” Following this statement, a researcher at the think tank Demos said, “This marks a turning point for YouTube. For the first time, it’s dealing not only with reputation damage but revenue damage.” [2]. To investigate this claim and how it relates to my hypothesis, I collected some data regarding the boycotting of YouTube.

Firstly, I found a graph from Business Insider showing the effect of the boycott on YouTube’s projected advertisement revenue in 2017 [1]. The graph, shown below, shows a project loss in YouTube advertisement revenue after the boycotting events. Albeit the loss in earnings does not appear to be much in comparison to Google’s total advertisement earnings, perspective is important here. The amount lost, $755 million dollars, is enough to impact the incomes of many YouTube channels.

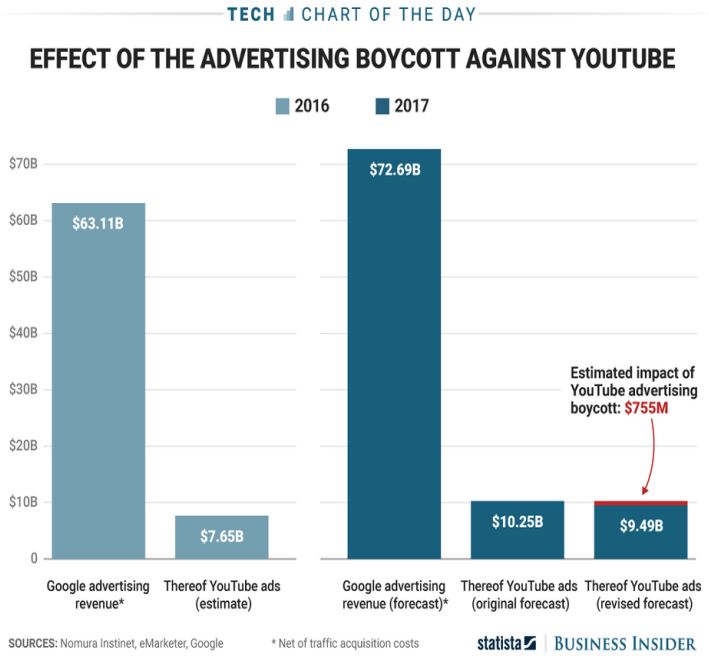

Fig. 2. Business Insider released a graph showing the impact of YouTube advertising boycott [1].

This loss not only discourages new content creators from joining the platform, but it forces existing content creators on YouTube to look for other sources of income (possibly day jobs). Of course, Google itself has not been affected by this change. I found some other data that shows how even after the “Adpocalypse” Google remains the number one benefactor of ad revenue on the Internet. This is an issue exclusively affecting the YouTube platform, which is precisely why YouTube should be considering making changes to their platform in response.

One could argue, that YouTube has already responded to this issue by tightening their monetization policies. They introduced the YPP, as mentioned earlier, and have allowed advertisers the option to filter content they do wish top sponsor. However, these changes are YouTube’s way of

enticing advertisers to come back to their platform. It has become much more difficult to monetize videos on YouTube with the changes made to their monetization algorithm. Now that YouTube has appeased the concerns of advertisers, they should look to support the content creators that are the basis of their platform.

Net Digital Ad Revenue Share Worldwide, by Company, 2016-2019
% of total and billions

	2016	2017	2018	2019
Google	32.8%	33.0%	32.4%	32.3%
Facebook	14.1%	16.2%	17.7%	18.7%
Alibaba*	6.6%	7.8%	9.0%	9.2%
Baidu	4.3%	4.2%	4.3%	4.5%
Tencent	2.2%	3.0%	4.0%	4.9%
Microsoft (Microsoft and LinkedIn)**	2.3%	2.7%	3.1%	3.5%
Yahoo	1.6%	1.4%	1.2%	1.1%
Twitter	1.2%	1.0%	0.9%	0.9%
Amazon	0.7%	0.8%	0.9%	1.0%
Verizon (AOL and Millennial Media)	0.7%	0.7%	0.6%	0.5%
Pandora	0.6%	0.5%	0.5%	0.5%
IAC	0.5%	0.4%	0.3%	0.3%
Sina	0.4%	0.4%	0.4%	0.5%
Snapchat	0.2%	0.4%	0.7%	1.1%
Sohu.com	0.5%	0.4%	0.4%	0.3%
Yelp	0.3%	0.3%	0.3%	0.3%
Other	31.1%	26.7%	23.2%	20.4%
Total digital ad spending (billions)	**$190.57**	**$223.74**	**$259.84**	**$297.41**

*Note: includes advertising that appears on desktop and laptop computers as well as mobile phones, tablets and other internet-connected devices, and includes all the various formats of advertising on those platforms; net ad revenues after company pays traffic acquisition costs (TAC) to partner sites; *includes ad revenue from Alibaba's core commerce operations and Youku Tudou; **includes ad revenue from LinkedIn*
Source: company reports; eMarketer, March 2017
224290 www.eMarketer.com

Fig. 3. Statistical breakdown of the ad revenue share across the world. Google remains the ad giant amongst tech companies [3].

Monetization has been an issue on YouTube even prior to the "Adpocalypse". I found some interesting data from 2014 that show how YouTube's demonetization algorithm demonetized a video for a content creator, but they were not notified [4].

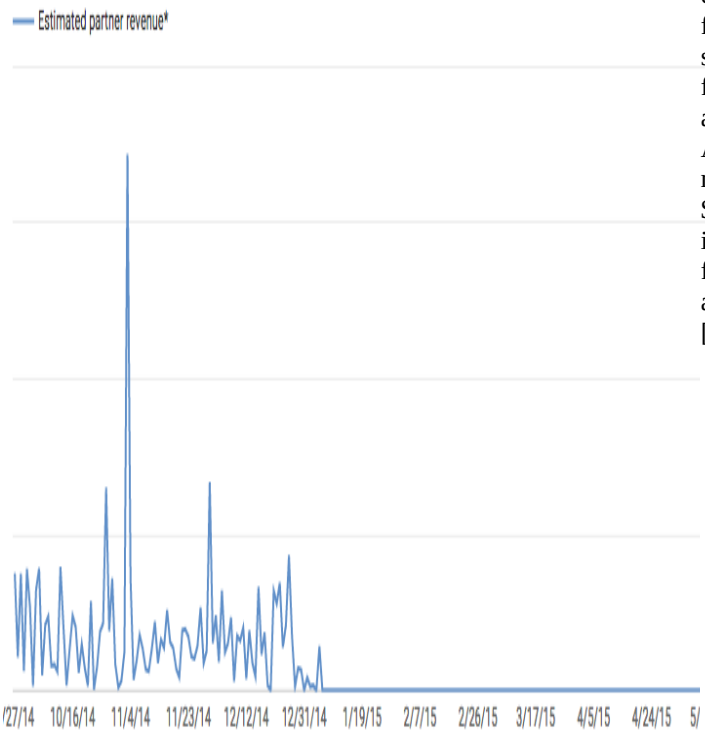

Fig. 4. Revenue graph for a video that was de-monetized in 2014. The creator did not know until Sept 2016 [4].

Clearly, monetization of videos cannot be a source of income for content creators anymore. Any video can become flagged by YouTube's generic filter list (mentioned above). Even though there is an appeals process to re-monetize videos, the process is tedious and often delays payments to content creators which can be hazardous for immediate bills. Not to mention, many of these advertisements are CPV and require users to watch the advertisement for them to become of any monetary value. With many software tools such as ad blockers prevalent on the Internet, advertisements should not be viewed as a stable source of income for YouTube content creators. YouTube must make a change to their monetary policy, but not by tweaking their advertisements algorithms. As we can see from the above data, content creators cannot and will not rely on advertisements for income. That said, what new features should YouTube implement on their platform?

B. What do other platforms do?

To understand what YouTube must change on their platform to improve monetization for content creators, I first researched what other similar platforms do. I first look at Twitch.tv, an incredibly popular live streaming platform.

I continued my data analysis by looking at Twitch's (live video streaming platform) monetary features. On Twitch, "streamers" can live video stream on their channels and earn money through several sources of income. The first and most notable source of income are subscriptions. Viewers may opt to pay a monthly fee of $4.99/month to channels of their choice as a form of support to the streamer. This fee is recurring and gives subscribers the benefit of ad-free streams, access to subscriber only chat, channel exclusive emotes and other benefits depending on the channel. About fifty percent of the subscription fee goes towards the streamer and the other half goes to Twitch as a convenience fee. The most popular streamer on Twitch (Ninja) earns about $500,000 per month from his subscribers [5]. According to an investment firm, namely, Piper Jaffray regular professional streamers can earn anywhere from $3000-$5000 per month from subscribers on average. An important detail about this statistic is that the subscription fee is a flat fee. It does not vary in monetary value like advertisements due with their different types (CPC/CPV) [6].

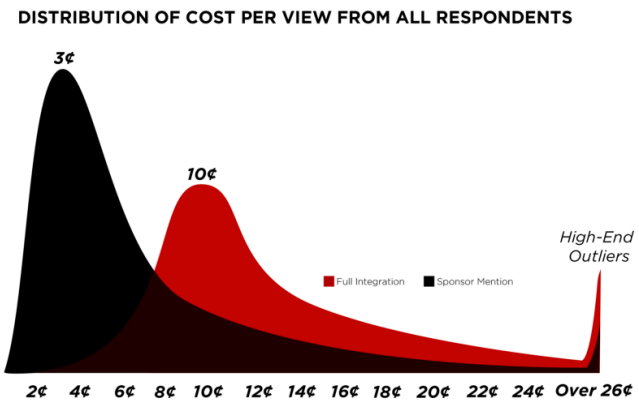

Fig. 5. Distribution of Cost Per View (CPV) revenue across several content creators [6].

The below distributions show just how much CPV revenues vary across content creators with sponsor mentions and those with full integrations with a brand. Twitch's consistency in payments is a very attractive feature that YouTube should try to implement on their platform.

Not only can streamers earn money through subscriptions on Twitch, but they can also monetize their content with advertisements, receive donations and sell merchandise. Streamers can make anywhere $0.9-$2 per 1000 views. They can also integrate a merchandise website into their channel and sell T-shirts with partnered T-shirt manufacturer Teespring [7]. What is important to note from this information, is the amount of variety Twitch provides their content creators to monetize their content. The myriad sources of income allow creators to expand their brand with ease, provided they are received interest from viewers.

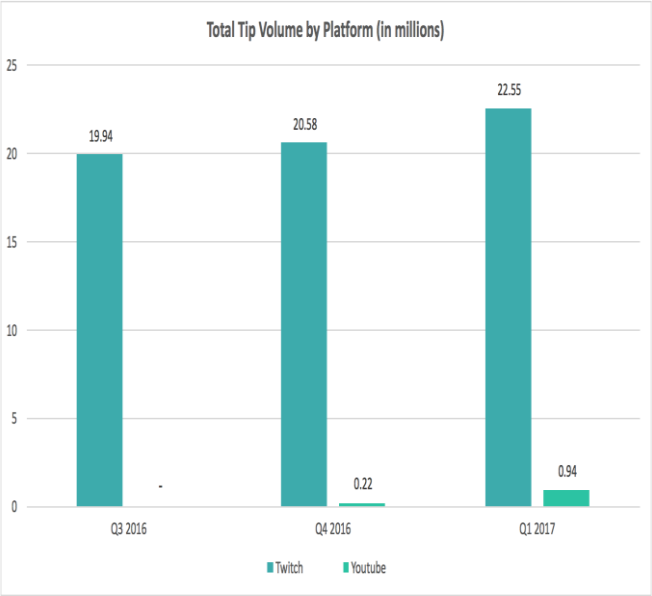

Fig. 6. Money earned from donations in Twitch and YouTube Live in Q3 & Q4 2016 and Q1 2017. Twitch is winning by a landslide [8].

The above statistics were taken from a 3rd party software called Streamlabs, which allows Twitch streamers to accept donations from viewers along with a message displayed on the stream. Streamers on YouTube Live can use this software as well, however, as evident from the above data they do not make nearly as much as Twitch streamers from these donations.

From this information, I inferred that YouTube can improve their monetization policies by introducing their own implementation of a donation feature. Currently YouTube channels only accept donations through another 3rd party application called Patreon. They link their Patreon account in video descriptions boxes and accept monthly paid subscriptions on Patreon with the benefit of exclusive content. In 2016, 35 content creators earned $150,000 on Patreon (including YouTubers) and a few content creators earned up to $450,000 in that year [11].

III. Concluding Results

After performing an analysis and comparison of platforms like YouTube, I have come to conclude that it would be in the best interest of YouTube to implement a donation or paid subscriber feature. It is evident that advertisements are no longer reliable sources of income and content creators should not have to depend on a 3rd party software like Patreon. YouTube should provide all the monetization tools necessary for content creators to grow. Like Twitch and Patreon, they can take convenience fee out of the donation transactions. YouTubers should not have to grow an audience on two platforms for the purposes of monetization. Such a monetization model will only deter new content creators from joining YouTube's platform. Of course, as per YouTube's policies this feature should only be available to partnered channels. An alternative solution would be to increase the convenience fee as the channel gains more subscribers and allow channels of reasonable size (1000 subscribers) to make use of the feature.

Here are some possible problems I anticipate will arise with this feature:

1. The feature will not be successful as YouTube's current model emphasizes free viewership.
2. YouTubers will not switch from 3rd party applications like Patreon, because YouTube's convenience fees will be too high.
3. The donation feature will continue to be exclusive to YouTube streaming or some other subset of YouTube as it has once been in the past.

To avoid the above potential problems with a donation feature, YouTube should strive to integrate donations in YouTube culture. On Twitch, donating is a very organic process as viewers have come to trust the secureness of transactions over Streamlabs. YouTube needs to build that trust with content creators and viewers to successfully implement a donation feature. They should also incentivize their new feature, by providing milestones for content creators. For example, after a 100 donations/100 paid subscribers a channel could be allowed higher video upload quality or a donator/subscriber only comment section. As for as the exclusivity of the new feature, it is ok if YouTube keeps the feature exclusive to certain channels while testing it. However, long term it is in YouTube's best interest to provide better monetization tools for their active content creators.

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